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Create a Simple Network using Cisco Packet Tracer

AIM:

To design and build a simple Local Area Network (LAN) in Cisco Packet Tracer by connecting end devices through a switch, assigning IP addresses, and verifying end-to-end connectivity with the ping command.

SOFTWARE REQUIRED:

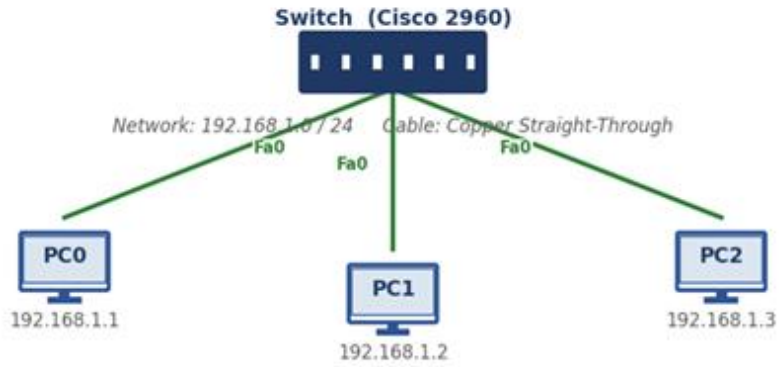
- A computer connected to a live network (Ethernet or Wi-Fi).
- Cisco Packet Tracer (version 8.x recommended
- Basic knowledge of IPv4 addressing and subnet masks.

THEORY:

Cisco Packet Tracer is a network simulation tool that lets you build and test networks without physical hardware. A simple LAN connects several end devices (PCs) to a central switch. A switch forwards frames using MAC addresses and creates one broadcast domain, so all devices attached to it — if given IP addresses in the same subnet — can communicate directly. In this experiment three PCs are joined to a single switch on the network 192.168.1.0/24 and connectivity is proven with ping, which sends ICMP echo requests and waits for replies.

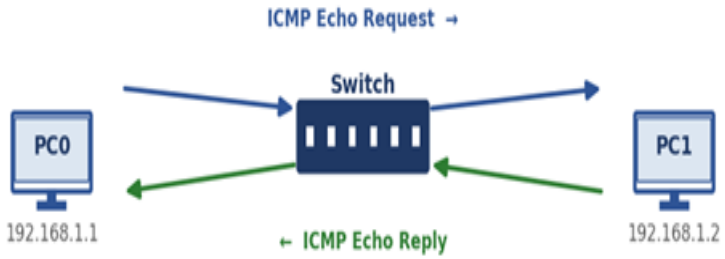
Addressing plan

Device	Interface	IP Address	Subnet Mask
PC0	FastEthernet0	192.168.1.1	255.255.255.0
PC1	FastEthernet0	192.168.1.2	255.255.255.0
PC2	FastEthernet0	192.168.1.3	255.255.255.0



PROCEDURE:

1. Open Cisco Packet Tracer. You are presented with a blank logical workspace (the large white area).
2. Add a switch: from the bottom-left device box choose **Network Devices → Switches**, pick the **2960** model, and click on the workspace to drop it.
3. Add three PCs: choose **End Devices**, select **PC**, and place PC0, PC1 and PC2 around the switch.
4. Connect the devices: choose the **Connections** (lightning-bolt) icon and pick **Copper Straight-Through** cable. Click a PC, choose its FastEthernet0 port, then click the switch and choose a free FastEthernet port. Repeat for all three PCs.
5. Wait until the link lights turn green (a few seconds), indicating the ports are up.
6. Configure IP addresses: click **PC0 → Desktop tab → IP Configuration**. Select Static and enter IP 192.168.1.1 and subnet mask 255.255.255.0. Close the window.
7. Repeat step 6 for PC1 (192.168.1.2) and PC2 (192.168.1.3), all with mask 255.255.255.0.
8. Test connectivity: on **PC0 → Desktop → Command Prompt**, run the ping command to another PC.
9. PC> ping 192.168.1.2
   PC> ping 192.168.1.3
10. A successful ping shows four replies with round-trip times. Repeat from other PCs to confirm every device can reach every other device.
11. Optional: switch to **Simulation mode** (bottom-right) and re-run a ping to watch the ICMP packet animate hop-by-hop from source to switch to destination.
12. Save the file via File → Save As with a .pkt extension.



SAMPLE OBSERVATION TABLE:

```
C:\> ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

If the first request “times out”, it is normal — it is caused by ARP resolving the MAC address; subsequent replies confirm the link is working.

RESULT:

Thus a simple LAN of three PCs connected through a 2960 switch was built in Cisco Packet Tracer. IP addresses were assigned in the 192.168.1.0/24 network, and end-to-end connectivity between all devices was successfully verified using the ping command.

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Study of Topology - Benchmark LAN Trainer Kit

AIM:

To study the various network topologies - bus, star, ring, mesh, tree and hybrid- and to physically construct and observe them using the Benchmark LAN Trainer Kit.

SOFTWARE REQUIRED:

- Benchmark LAN Trainer Kit with its node modules, central hub/switch module and patch cords.
- Set of connecting cables (RJ-45 patch cords or the kit's supplied connectors).
- Kit power supply and the accompanying observation/indicator panel.

THEORY:

Network topology is the physical or logical arrangement in which nodes (computers, terminals or the kit's node modules) are interconnected. The topology determines how data flows, how much cabling is needed, how the network behaves when a link or node fails, and how easy it is to expand. A LAN trainer kit provides node modules and a central connecting module so that each topology can be wired up by hand and its behaviour — signal path, fault effect, expandability — directly observed on the kit's indicator panel.

Addressing plan

Topology	Arrangement	Advantages	Disadvantages
Bus	All nodes share one common backbone cable with terminators at both ends.	Simple; least cabling; cheap.	One cable fault stops the whole network; hard to troubleshoot.
Star	Every node connects to a central hub/switch.	Easy to add/remove nodes; single node fault does not affect others.	Central device failure downs the whole network; more cable.
Ring	Nodes form a closed loop; data passes node to node.	Orderly, predictable data flow; no collisions with token passing.	A single break can halt the ring; adding nodes disrupts it.
Mesh	Every node links directly to every other node.	Highly reliable; multiple paths; fault-tolerant.	Very expensive; huge amount of cabling.
Tree	Groups of star networks joined to a bus backbone (hierarchy).	Scalable; good for large structured networks.	Backbone failure affects large sections; complex.
Hybrid	Combination of two or more of the above.	Flexible; tailored to needs.	Design and management are complex and costly.

PROCEDURE:

1. Place the LAN Trainer Kit on the bench and connect its power supply. Switch it on and confirm the indicator lamps/LEDs are functioning.
2. Identify the components: the individual node modules, the central hub/switch module, the backbone connector, and the supplied patch cords.

Bus topology

3. Connect all node modules along the single common backbone line provided on the kit, fitting the terminators at both ends of the bus.
4. Send data from one node and observe on the indicator panel that the signal travels along the shared backbone and is seen by all nodes; only the addressed node accepts it.
5. Disconnect the backbone at a mid-point and note that communication beyond the break stops — demonstrating the bus's single point of failure.

Star topology

6. Connect each node module by a separate patch cord to a port on the central hub/switch module.
7. Transmit from one node and observe that data passes through the central module to reach the destination node.
8. Disconnect one node's cable and confirm the other nodes continue to communicate; then power off the central module and note that the whole network stops — showing the star's dependence on the central device.

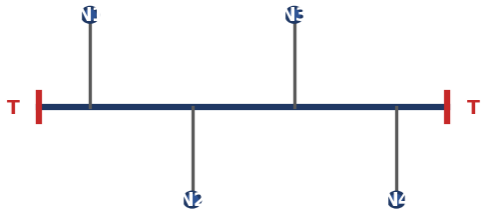
Ring topology

9. Wire the node modules in a closed loop — the output of each node to the input of the next — until the last node connects back to the first.
10. Send data and observe it circulating from node to node around the ring until it reaches the intended destination.
11. Open the loop at one link and observe that the ring's data flow is interrupted.

Mesh topology

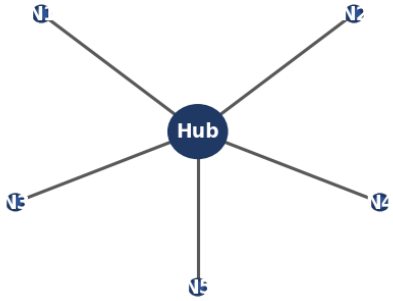
12. Using the available patch cords, connect each node module directly to every other node module.
13. Transmit between two nodes and observe that, because multiple direct paths exist, communication continues even if one link is removed — demonstrating fault tolerance.
14. For each topology, record on the observation sheet: number of cables used, the path taken by the data, and the effect of a single link or node failure.

(a) Bus Topology



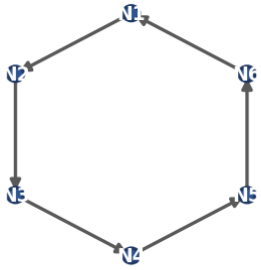
Common backbone + terminators

(b) Star Topology



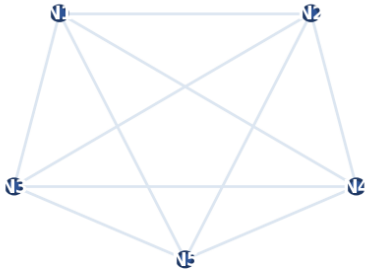
All nodes → central hub / switch

(c) Ring Topology



Closed loop, data passes node to node

(d) Mesh Topology



Every node linked to every other node

SAMPLE OBSERVATION TABLE:

Topology	Cables used (for n nodes)	Effect of one link failure	Ease of adding a node
Bus	n + backbone	Whole segment may fail	Easy
Star	n	Only that node affected	Very easy
Ring	n	Ring flow interrupted	Difficult
Mesh	n(n−1)/2	No effect (alternate paths)	Difficult

RESULT:

Thus, the bus, star, ring and mesh network topologies were studied and physically constructed on the Benchmark LAN Trainer Kit.